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Integer Quantization and Structured Pruning for Sub-Milliwatt Edge Deep Neural Networks

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ABSTRACT

Deploying modern deep neural networks on energy-harvesting microcontroller units requires extreme model compression without catastrophic accuracy degradation. We introduce an iterative post-training quantization and structured kernel pruning pipeline that compresses 32-bit floating-point models to 4-bit integer representations. Our approach uses Hessian-weighted sensitivity metrics to preserve salient attention weights while pruning 68% of redundant convolutional filter blocks. The resulting compressed models operate under a 0.8 milliwatt power envelope with only 1.2% top-1 accuracy loss across standard benchmark datasets. However, dynamic activation ranges in highly non-linear layers occasionally cause integer overflow clipping during out-of-distribution inputs.

Keywords— Machine Learning, Model Compression, Quantization, Edge Inference

INTRODUCTION

Edge intelligence requires neural networks to operate within micro-watt power budgets on battery-free microcontrollers. Existing pruning algorithms often generate unstructured sparse matrices that cannot be accelerated on simple integer hardware.

METHODOLOGY

We combine second-order Hessian sensitivity analysis with symmetric 4-bit integer uniform quantization. Filter blocks with minimal curvature contribution are pruned, leaving dense structured integer sub-matrices optimized for SIMD execution.

RESULTS

Tested on ARM Cortex-M55 and ESP32 microcontrollers, our compressed vision and audio models ran at 45 FPS drawing only 0.78 mW, representing a 9.2x efficiency gain over 8-bit quantized baselines.

LIMITATIONS

Extreme 4-bit quantization reduces dynamic range; when exposed to abrupt sensory spikes or out-of-distribution inputs, accumulator registers can saturate, causing temporary accuracy dips.

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